

Textual Reasoning Attention Map

Ten distinct books are to be distributed among three students: Alice, Bob, and Charlie. Each student must receive at least one book. The number of books Alice receives must be strictly greater than the number of books Bob receives, and the number of books Bob receives must be strictly greater than the number of books Charlie receives. Let N be the number of ways to distribute the books satisfying these conditions. Find the remainder when N is divided by 1000.

The number of ways to distribute 10 distinct books among Alice, Bob, and Charlie such that each receives at least one book and Alice > Bob > Charlie (strictly) is found by enumerating all positive integer triples (a, b, c) with $a + b + c = 10$ and $a > b > c$. These triples are $(7, 2, 1)$, $(6, 3, 1)$, $(5, 4, 1)$, and $(5, 3, 2)$. For each triple, the number of assignments of books is $\frac{10!}{a! b! c!}$. Computing these: $(7, 2, 1)$: $\frac{10!}{7! 2! 1!} = 360$. $(6, 3, 1)$: $\frac{10!}{6! 3! 1!} = 840$. $(5, 4, 1)$: $\frac{10!}{5! 4! 1!} = 1260$. $(5, 3, 2)$: $\frac{10!}{5! 3! 2!} = 2520$. Summing gives $N = 360 + 840 + 1260 + 2520 = 4980$. The remainder when 4980 is divided by 1000 is 980. Thus, the answer is 980.

Visual Reasoning Attention Map

Ten distinct books are to be distributed among three students: Alice, Bob, and Charlie. Each student must receive at least one book. The number of books Alice receives must be strictly greater than the number of books Bob receives, and the number of books Bob receives must be strictly greater than the number of books Charlie receives. Let N be the number of ways to distribute the books satisfying these conditions. Find the remainder when N is divided by 1000.

The number of ways to distribute 10 distinct books among Alice, Bob, and Charlie such that each receives at least one book and Alice > Bob > Charlie (strictly) is found by enumerating all positive integer triples (a, b, c) with $a + b + c = 10$ and $a > b > c$. These triples are $(7, 2, 1)$, $(6, 3, 1)$, $(5, 4, 1)$, and $(5, 3, 2)$. For each triple, the number of assignments of books is $\frac{10!}{a! b! c!}$. Computing these: - $(7, 2, 1)$: $\frac{10!}{7! 2! 1!} = 360$ - $(6, 3, 1)$: $\frac{10!}{6! 3! 1!} = 840$ - $(5, 4, 1)$: $\frac{10!}{5! 4! 1!} = 1260$ - $(5, 3, 2)$: $\frac{10!}{5! 3! 2!} = 2520$ Summing gives $N = 360 + 840 + 1260 + 2520 = 4980$. The remainder when 4980 is divided by 1000 is 980. Thus, the answer is 980.

High

Low

Attention Intensity